

Response to the Editors and Reviewers

IEEE Transactions on Circuits and Systems for Video Technology

Manuscript TCSVT-29693-2026: *Beyond Homogeneous Adaptation: Decoupled Control for Sample- and Layer-Aware Test-Time Robustness in Temporally Correlated Streams*

Dear Dr. Shan Liu (Editor-in-Chief), Associate Editor, Senior Area Editor, and Reviewers,

Thank you for your careful evaluation and constructive suggestions. We appreciate the reviewers' recognition of the motivation, technical design, and long-horizon evidence of the Decoupled Control Framework (DCF). Your comments helped us strengthen the analytical interpretation, experimental validation, computational cost assessment, and reproducibility of DCF.

We summarize the main revisions in three groups below, then respond to each comment and identify the corresponding manuscript changes.

1. Probe interpretation and layer behavior. We strengthened the local directional-sensitivity interpretation of PCS and added controlled shape–texture and Colored-MNIST shortcut experiments, matched-coverage routing diagnostics, and frequency/strength sensitivity analysis. These experiments connect probe responsiveness to reliable routing under the evaluated conditions. We also added retention-gate heatmaps over time and corruption types, together with control-granularity and Expected Fisher/Hutchinson-based proxy comparisons.

2. Geometry repair and computational cost. We compared RGR with moment matching, prototype contrastive alignment, and uniform-prior OT, and characterized failure regimes through 10%–90% routing-ratio tests. We report FLOPs, peak memory, and latency alongside lightweight baselines. On the measured A100 setup, DCF-Lite reduces FLOPs by 61.8% and peak memory by 61.3%, achieving 42.06% accuracy versus 43.48% for full DCF. These results quantify an accuracy–resource trade-off; feasibility on a specific edge device requires device-level evaluation.

3. Statistical evidence, positioning, and reproducibility. Five matched runs and paired tests support significant aggregate cross-domain transfer improvements on ImageNet-C ($p = 1.39 \times 10^{-4}$) and DomainNet-126 ($p = 2.75 \times 10^{-4}$). We clarify the trainable parameters, CLR grouping, routing regions, threshold guidance, and OT settings. We also discuss four closely related TCSVT studies ([29], [30], [31], [36]), the suggested sensor-stream literature ([19]–[21]), and recent continual-TTA work ([28], [38]).

Below, we respond to each editorial request and every numbered reviewer comment. Original comments appear in shaded boxes, responses follow directly, and selected manuscript additions appear in blue boxes with specific locations. Accuracy differences are expressed in percentage points (pp).

Sincerely,

Zhengpeng Zhao and Yuanyuan Pu

Corresponding authors, on behalf of all authors

Summary of Major Revisions and Roadmap of Itemized Responses

Reviewer / Editor	Core topic / request	Key revision / action	Manuscript location
Editor-in-Chief (E1)	TCSVT related papers	Identified four closely related TCSVT papers and provided direct mechanism-level distinctions from DCF.	Section II, p. 3; Refs. [29]–[31], [36]
Editor-in-Chief (E2)	Distinctive contribution	Clarified DCF as coordinated route–adapt–retain control of a coupled sample–layer feedback process.	Sections I–II, III-B, pp. 1–3
Associate Editor (AE1–AE5)	Major-revision synthesis	Addressed probe grounding/sensitivity, RGR alternatives/failure cases, cost, CLR behavior, and statistics.	Sections III–IV, pp. 4–12
Senior Area Editor	Ablation depth	Expanded component ablations into mechanism-level diagnostics and efficiency/transfer analyses.	Section IV-C, pp. 8–12
Reviewer 1 (R1.1)	Fourier probe grounding	Clarified the local PCS interpretation and added controlled shape–texture/color shortcuts, matched-coverage purity diagnostics, and frequency/strength sensitivity analysis.	Sec. III-C; IV-C; Tables IV–VI; Fig. 12(c)
Reviewer 1 (R1.2)	RGR alternatives/failure modes	Compared simpler alternatives and fixed/dynamic priors; added 10–90% routed-away stress test and safeguards.	Sec. III-D; IV-C; Tables VII–VIII
Reviewer 1 (R1.3)	Computational overhead	Reported FLOPs, peak memory, latency, throughput context, and DCF-Lite approximations.	Sec. IV-C; Table XI; Fig. 12(d)
Reviewer 1 (R1.4)	CLR layer behavior	Added retention-gate heatmaps, control-granularity comparison, Expected Fisher, and Hutchinson approximation.	Sec. III-E; IV-C; Fig. 9; Table IX
Reviewer 1 (R1.5)	Suggested literature	Integrated the three suggested uncertainty-aware SHM studies.	Section II, p. 3; Refs. [19]–[21]
Reviewer 1 (R1.6)	Statistics and broader transfer	Added five-run variability, paired tests, and DomainNet-126 transfer.	Section IV-C; Fig. 11

Reviewer / Editor	Core topic / request	Key revision / action	Manuscript location
Reviewer 2 (R2.1)	Trainable parameters/layer definition	Specified normalization affine parameters and ResNet/ViT CLR grouping; compared granularities.	Section IV-A; Table IX
Reviewer 2 (R2.2)	Four routing regions	Defined Areas 1–4 with explicit entropy/PCS inequalities.	Section III-C; Fig. 7
Reviewer 2 (R2.3)	Threshold/OT guidance	Added practical threshold rule and full Sinkhorn/OT settings.	Section IV-A; Fig. 12(b),(d)
Reviewer 2 (R2.4)	Recent continual TTA	Added GOLD (CVPR 2026) and the 2026 continual-TTA survey, with direct comparison to DCF.	Section II; Refs. [28], [38]

Note: section, figure, table, equation, reference, and manuscript page numbers below refer to the revised manuscript unless stated otherwise.

Response to the Editor-in-Chief

E1 Closely related papers published in TCSVT

Comment

Before your new submission, you have to answer two questions clearly in your revised manuscript and responses: a) what are the 3-5 papers published in the IEEE Transactions on Circuits and Systems for Video Technology, which are most closely related to your manuscript; b) what is distinctive / new about your current manuscript related to these previously published papers.

Response

Section II now discusses the following four closely related TCSVT studies. For each study, we explain how its adaptation mechanism relates to DCF and where DCF differs.

[31] J. Liu, J. Xie, F. Zhou, and S. He, “Question Type-Aware Debiasing for Test-Time Visual Question Answering Model Adaptation,” IEEE TCSVT, vol. 34, no. 11, pp. 10805–10816, 2024.

QED addresses unreliable test-time supervision in VQA by using question-type-aware entropy and negative perturbations to identify biased samples. Both QED and DCF address the reliability of samples used for adaptation. DCF extends reliability control across the full adaptation loop: it combines confidence with a structured Fourier stress response to route target evidence, reuses routed-away samples through soft optimal-transport geometry repair, and selectively retains the resulting model update across layer groups.

[29] J. Liu and Z. Yang, “Test-Time Adaptation for Real-World Video Adverse Weather Restoration With Meta Batch Normalization,” IEEE TCSVT, vol. 35, no. 6, pp. 5533–5544, 2025.

MetaBN improves adverse-weather video restoration through source-stage meta-learning of normalization behavior and test-time self-supervised updates. MetaBN focuses on rapid task-specific adaptation through meta-trained BN parameters. DCF starts from an ordinary pre-trained recognition model and directly controls the reliability and persistence of online target-driven updates in temporally correlated streams. The source model provides the reference for curvature-aware retention throughout online adaptation.

[30] C. Zhang et al., “Collaborative Model and Data Adaptation at Test Time,” IEEE TCSVT, vol. 36, no. 6, pp. 8980–8994, 2026.

CMDA jointly adapts the model and the test data, using diffusion-based data adaptation to move target samples toward the source distribution while updating the model. DCF adapts directly to the incoming target stream through three coordinated decisions: sample routing determines which evidence may drive consistency learning, routed-away samples preserve target geometry through OT, and CLR decides layer by layer how much of the candidate update is retained.

[36] J. Han et al., “Unleashing the Potential of All Test Samples: Mean-Shift Guided Test-Time Adaptation,” IEEE TCSVT, 2026.

MS-TTA is especially relevant to our treatment of samples that are not suitable for direct confident supervision. It refines all CLIP test features using training-free mean shift and a cache, thereby extracting value from low-confidence samples. DCF assigns different adaptation roles to the two subsets: trusted samples provide direct consistency supervision, whereas routed-away samples contribute only through soft class-geometry repair. This separation prevents weak evidence from entering the label-like objective while still exploiting its distributional structure; CLR then controls whether the resulting update should persist in each layer group.

These studies address complementary aspects of reliable, efficient, and stable test-time adaptation. DCF advances this line of work by coupling three control decisions that are otherwise handled separately: *which target evidence is admitted to direct adaptation, how non-admitted evidence is still exploited, and where in the network the resulting update is allowed to persist*. This coordinated route–adapt–retain control directly targets the error-amplifying feedback loop that emerges in temporally correlated streams.

Changes in the manuscript: Section II, “Related Work,” p. 3; references [29], [30], [31], and [36].

E2 Distinctive contribution relative to these TCSVT studies

Response

The central novelty of DCF is a unified route–adapt–retain control architecture for the coupled sample–layer dynamics of online TTA. The TCSVT studies above emphasize sample reliability, normalization behavior, data adaptation, or feature refinement. DCF links three decisions in one online loop: PSR selects evidence for direct consistency supervision; RGR reuses routed-away samples as target-geometry constraints; and CLR treats the sample-side update as a candidate whose persistence is determined separately for different layer groups. By linking these decisions, DCF controls reliability and stability throughout the online adaptation loop.

This coupling is important under temporally correlated streams because an erroneous target update can simultaneously distort the evidence used by later samples and accumulate unevenly across network depth. DCF directly controls both sides of this feedback process. The component and mechanism-level ablations in Section IV-C isolate the gains from routing, geometry repair, and layer-wise retention, while the long-horizon experiments show that their coordination prevents the late-stage collapse observed in multiple advanced TTA baselines.

Selected revised text

“DCF instead models their interaction as a coupled sample–layer feedback process, jointly determining which samples drive adaptation, how routed-away samples preserve target structure, and which layers retain the resulting updates.”

Changes in the manuscript: Sections I–II, pp. 1–3; Section III-B, p. 3, Eqs. (2)–(3); Section IV-C, Table III and the mechanism-level analyses, pp. 8–12.

Response to the Associate Editor

Thank you for identifying the analyses needed for this major revision. We address the five substantive concerns below; the corresponding Reviewer 1 responses provide the detailed evidence.

AE1 Justification and sensitivity of the Fourier probe

Comment

The Fourier-based stress probe lacks formal theoretical justification, and a sensitivity study on the choice of frequency range and perturbation strength is needed.

Response

We now explain the Fourier probe using a local directional-sensitivity analysis and controlled experiments. PCS is the positive drop in the original predicted-class probability; Response R1.1 below makes this positive-part operation explicit in its local expansion. The analysis explains what PCS measures. Shape–texture interventions, paired shortcut counterfactuals, matched-coverage purity analysis, and Colored-MNIST then test how that response relates to routing reliability. Figure 12(c) further evaluates frequency range and perturbation strength around the default setting.

Changes in the manuscript: Section III-C, p. 4; Section IV-C, Tables IV–VI, pp. 8–9, and Fig. 12(c), p. 12. See Response R1.1.

AE2 Alternatives and failure cases for geometry repair

Comment

The routed geometry repair module should be compared against simpler alternatives, and potential failure cases when the routed set is small or the prior estimate is noisy should be discussed.

Response

We added the requested alternatives and tested how RGR behaves as the proportion of routed-away samples changes. Table VII compares discarding routed-away samples, feature moment matching, prototype contrastive alignment, fixed-uniform-prior OT, and the proposed dynamic-prior OT while keeping the remaining DCF components unchanged. We further added a routing-ratio stress test from 10% to 90% routed-away samples and clarified the prior EMA, uniform interpolation, centroid momentum, and empty-set safeguards. These tests show stable performance when enough trusted samples remain, and a clear decline when routed-away samples dominate the batch.

Changes in the manuscript: Section III-D, pp. 4–5; Section IV-C, Tables VII–VIII, p. 10; Algorithm 1, p. 6. See Response R1.2.

AE3 Computational cost and practical approximations

Comment

Computational overhead, including FLOPs and memory, should be reported in detail alongside a discussion of possible approximations for practical deployment.

Response

Table XI now provides a complete resource comparison for DCF and representative baselines, including FLOPs, peak GPU memory, per-image latency, and accuracy under the same A100 evaluation setting. Full DCF achieves 43.48% long-horizon average accuracy with 24.6 GFLOPs, 14,690 MB peak memory, and 5.20 ms/image. DCF-Lite reduces the cost to 9.4 GFLOPs and 5,680 MB with 3.59 ms/image while retaining 42.06% accuracy, corresponding to reductions of 61.8% in FLOPs and 61.3% in peak memory for only a 1.42 pp accuracy decrease. Figure 12(d) and Table IX further quantify low-iteration Sinkhorn and curvature-proxy approximations, making the accuracy–efficiency trade-off explicit.

Changes in the manuscript: Section IV-C, “Computational Overhead,” pp. 11–12; Table XI and Fig. 12(d), p. 12. See Response R1.3.

AE4 Behavior of the layer retention gates

Comment

The per-layer behaviour of the curvature-aware retention mechanism needs further analysis, for example through a heatmap of retention gates over time and corruption types.

Response

Figure 9 now shows the layer-wise retention gates throughout the 225-domain stream and across corruption types. The gates vary with both network depth and corruption type: layer4 remains highly permissive to candidate updates, while earlier and intermediate groups show stronger temporary source anchoring under specific corruptions. The temporal trajectories recover after transient suppression, demonstrating that CLR preserves adaptation capacity through dynamic retention. Table IX further compares control granularities and alternative curvature estimators.

Changes in the manuscript: Section III-E, pp. 5–6; Section IV-C, Fig. 9 and Table IX, p. 10. See Response R1.4.

AE5 Statistical evidence for cross domain transfer

Comment

Additionally, statistical significance should be reported for the cross-domain results, where some improvements are marginal.

Response

Figure 11 now reports mean $\pm$ standard deviation over five matched runs and paired significance tests against the strongest aggregate baseline in each transfer setting. On ImageNet-C, DCF achieves 30.30 ± 0.39 pp versus 27.95 ± 0.05 pp for AEA, with $t(4) = 14.30$ and $p = 1.39 \times 10^{-4}$. On DomainNet-126, DCF achieves 61.40% versus 60.13% for DeYO, with $t(4) = 12.01$ and $p = 2.75 \times 10^{-4}$. These paired tests establish statistically significant aggregate differences across matched runs; individual transfer cells are reported with their run-to-run variability.

Changes in the manuscript: Section IV-C, “Cross-Domain Transfer,” and Fig. 11, p. 11. See Response R1.6.

Response to the Senior Area Editor

Comment

The paper proposes the use of a decoupled control system for reducing bias in target evidence in test-time adaptation using online methods for temporally correlated image streams. The topic is relevant to TCSVT. The paper is complete with contributions, experiments, results, and a limited ablation study.

Response

Thank you for recognizing the relevance and completeness of the work. To address the limited ablation study, we expanded Section IV-C with controlled PSR diagnostics; RGR alternatives, prior ablation, and routing-ratio stress tests; CLR gate visualizations, granularity and proxy comparisons; cross-domain statistics; and DCF-Lite efficiency analysis. These additions clarify how the control mechanisms operate and where their limits arise.

Changes in the manuscript: Section IV-C, Tables IV–IX and XI, pp. 8–12; Figs. 9, 11, and 12, pp. 10–12.

Response to Reviewer 1

We appreciate your positive assessment of the motivation and long-horizon results. Your comments motivated additional analytical grounding, controlled shortcut experiments, mechanism-level robustness studies, and deployment-oriented evaluation. We address the six comments individually below.

R1.1 Fourier probe justification and sensitivity

Comment

The paper empirically shows that the Fourier perturbation exposes stress-inert predictions, but lacks a formal justification of why frequency-domain stress specifically targets shortcut cues (e.g., texture biases) rather than semantic features. A theoretical analysis—or at least a controlled synthetic experiment where shortcuts are known (e.g., color vs. shape)—would strengthen the claim. Currently, the probe design (random frequency and direction) appears heuristic; a sensitivity study on the choice of frequency range and perturbation strength λ would also help practitioners set these parameters without extensive tuning.

Response

Thank you for this suggestion. We added a local sensitivity analysis, experiments with known shortcut factors, matched-coverage routing diagnostics, and a broader sensitivity study to explain and test the Fourier probe.

Local squared-response interpretation of PCS

Section III-C defines PCS as the positive drop in the probability assigned to the original predicted class under the stressed view. For a fixed input and original predicted class, let δ be the actual perturbation after clamping and g_x the input gradient of that class probability. To make the positive-part operation in Eq. (5) explicit, the local squared-response expansion is

$$\mathbb{E}_q[s(x)^2] = \mathbb{E}_q\left[\left(-g_x^\top \delta\right)_+^2\right] + O(\mathbb{E}_q\|\delta\|_2^3).$$

Thus PCS measures one-sided probability sensitivity along the directions excited by the Fourier probe. Let $Q_q(x) = \mathbb{E}_q[\delta\delta^\top]$ denote the perturbation second-moment matrix. The untruncated quadratic response is $g_x^\top Q_q(x)g_x$; for the positive-part PCS, the leading term retains the truncation and is bounded above by this quadratic response. No zero-mean or symmetry assumption is imposed on the perturbation after clamping.

Under a locally smooth predicted-class probability and sufficiently small perturbations, this expansion characterizes the directional sensitivity measured by PCS. The controlled interventions establish its connection to shortcut dependence: among

confidence-matched candidates in the evaluated settings, task-relevant predictions exhibit stronger PCS responses and are preferentially retained by the joint entropy–PCS rule. Together, the analytical interpretation and controlled experiments explain why PCS complements confidence in selecting reliable adaptation evidence.

Controlled shortcut evidence

We added a 224×224 synthetic benchmark in which circle/square shape is the task factor and four sinusoidal textures provide explicitly controlled shortcuts. Table IV reports both shape-decision agreement and separate shape/texture PCS responses, allowing us to measure differential probe sensitivity while verifying that the semantic shape decision is preserved.

Evidence from revised Table IV

Probe	Shape agreement (%)	Shape PCS ¹	Texture PCS ¹	PCS ratio
Lower nominal band	99.40	241.0	69.0	$3.49\times$
Middle nominal band	99.60	224.0	62.0	$3.61\times$
Upper nominal band	99.70	217.0	58.0	$3.74\times$
Energy-matched pixel noise	99.80	112.0	98.0	$1.14\times$
Random Fourier probe	99.60	229.0	64.0	$3.58\times$

¹ PCS entries are in units of 10^{-3} . The three nominal bands are [1, 75), [75, 150), and [150, 224] cycles/image. “Nominal” denotes the frequency sampling parameter before image-grid discretization.

Across preserved, randomized, and reversed shortcut correlations, PCS achieves 0.81 AUROC for distinguishing task-relevant from shortcut-driven predictions, compared with 0.71 for energy-matched pixel noise. At matched trusted-set coverage, shortcut contamination decreases from 24.6% to 11.8%. On ImageNet-C, Table VI further compares filters at the same frozen model state and matched coverage: pseudo-label purity is 49.9% for entropy + PCS, versus 44.0% for entropy alone and 47.1% for PCS alone, showing that PCS adds discriminative information beyond confidence alone. These results show that, under the evaluated settings, PCS provides information complementary to predictive entropy and enriches the trusted set for more reliable adaptation evidence.

We also added binary Colored-MNIST as a distinct color-shortcut setting. Digit identity defines the semantic target; training labels contain 25% noise, while test labels are clean and color–label agreement is approximately 30%. DCF obtains 88.91% accuracy, compared with 62.37% for DeYO and 50.93% for the source model.

Additional evidence from revised Tables V–VI

Diagnostic	Comparator / reference	DCF / joint-routing result
Colored-MNIST accuracy (Table V)	Best listed baseline: DeYO 62.37%	88.91% (+26.54 pp)
Pseudo-label purity at matched coverage (Table VI)	Entropy only 44.0%; PCS only 47.1%	Entropy + PCS 49.9%
Shortcut contamination at matched coverage	Confidence-matched candidates: 24.6%	Trusted-set contamination: 11.8%

Frequency range and perturbation strength

Figure 12(c) jointly varies the nominal frequency sampling range and perturbation strength λ . Section III-C specifies independent per-channel frequency and direction sampling, RMS-normalized Fourier bases, and exponential amplitude sampling with rate $1/\lambda$; the default is $\lambda = 0.2$ with nominal frequencies sampled from [1, 224] cycles/image. Performance remains stable across a broad neighborhood of this configuration, supporting the reported setting as a practical default with coarse parameter selection.

Selected revised text

“Thus, this relation establishes PCS as a structured directional-sensitivity measure whose response depends jointly on the prediction gradient and the directions excited by the Fourier probe.”

Changes in the manuscript: Section III-C, p. 4, Eqs. (4)–(7); Section IV-C, “Analysis of Probe-Supported Sample Routing,” pp. 7–9; Tables IV–VI, pp. 8–9; Fig. 12(c), p. 12.

R1.2 Geometry repair alternatives and failure modes

Comment

RGR uses entropic OT with dynamic class priors to align routed-away features to source-initialized centroids. However, the paper does not compare this with simpler alternatives like feature moment matching or contrastive learning. Moreover, the centroid update (Eq. 19) uses only routed-away samples; if the routed set dominates, centroids may drift away from source semantics. The authors should discuss failure cases when the routed set is small or when the prior estimate is noisy (e.g., early adaptation stages). Adding an ablation with fixed uniform priors would isolate the benefit of dynamic priors.

Response

Thank you for suggesting these comparisons. We added them to separate the effects of OT-based geometry repair and the dynamic prior, and used a routing-ratio stress test to examine when RGR remains effective.

Comparison with simpler alternatives

Table VII keeps PSR and CLR fixed and changes only the treatment of the routed-away subset. It compares discarding those samples, feature moment matching, prototype contrastive alignment, OT with a fixed uniform prior, and OT with the proposed dynamic prior.

Evidence from revised Table VII

Treatment of routed-away samples	T-CS (%)	T-CS-LS (%)	Long-horizon final (%)
Discard samples	41.87	42.71	42.88
Moment matching	42.81	42.82	41.51
Prototype contrastive alignment	42.83	42.79	42.33
OT with fixed uniform prior	43.01	42.80	42.91
OT with dynamic prior	43.49	43.29	43.84

The long-horizon final accuracy is 43.84% for dynamic-prior OT, compared with 42.91% for uniform-prior OT, a 0.93 pp improvement. Moment matching and contrastive alignment obtain 41.51% and 42.33%, respectively, versus 42.88% when routed-away samples are discarded. Thus, OT-based repair provides the strongest long-horizon result among the evaluated treatments, and the dynamic marginal contributes a further 0.93 pp gain over uniform-prior OT.

Conservative prior and centroid updates

Section III-D now spells out the prior construction. The batch prior uses stop-gradient predictions from the full batch, and its EMA starts from the uniform distribution with update rate $\rho = 0.01$. The OT class marginal then mixes this estimate with the uniform prior using $\alpha_b = 0.01$. Consequently, its L1 deviation from uniform is bounded by 0.02 and every class retains marginal mass of at least $0.99/K$, tightly regularizing transport allocation while still allowing target evidence to influence the class marginal.

The centroids are initialized from normalized rows of the frozen source classifier and updated outside backpropagation using soft OT-weighted feature means with momentum $\rho_M = 0.99$, followed by normalization. If the numerical transport mass of a class is negligible, its previous centroid is retained. Source initialization and slow momentum anchor early geometry while allowing gradual target adaptation. Persistent target bias can still induce drift, motivating the routing-ratio stress test below.

Small and dominant routed-away subsets

The revised method also handles empty subsets explicitly. When the routed-away subset is empty, the geometry loss is zero and the centroids remain unchanged. When the trusted subset is empty, the entire adaptation step is skipped and the model, centroids, and prior state are retained. The routing-ratio stress test examines how performance changes as the balance between trusted supervision and routed-away geometry information shifts.

Evidence from revised Table VIII

Routed-away proportion	T-CS accuracy (%)	Centroid drift
Native routing	43.49	0.0518

Routed-away proportion	T-CS accuracy (%)	Centroid drift
10%	42.98	0.0052
25%	43.49	0.0217
50%	43.18	0.0829
75%	41.64	0.1268
90%	22.12	0.0468

The controlled 10%–50% routed-away regime yields 42.98%–43.49% accuracy. Performance decreases to 41.64% at 75% and to 22.12% at 90%, showing that a dominant routed-away subset leaves insufficient trusted evidence for consistency learning. Centroid drift is smaller at 90% than at 75% despite the sharper accuracy loss, identifying trusted-supervision scarcity as a distinct failure mechanism. These results show the range of routing proportions over which RGR remains effective in this test and where its performance deteriorates.

Selected revised text

“The slow EMA mitigates early fluctuations, though persistent bias may affect the estimated prior. A tiny routed-away set limits target support, while an overly dominant one reduces trusted supervision.”

Changes in the manuscript: Section III-D, pp. 4–5, Eqs. (10)–(16); Algorithm 1, p. 6; Section IV-C, “Analysis of Routed-away Geometry Repair,” pp. 9–10; Tables VII–VIII, p. 10. The centroid update referred to as Eq. (19) in the original review is Eq. (16) in the revision.

R1.3 Computational overhead and lightweight deployment

Comment

DCF involves Sinkhorn iterations for OT, curvature estimation per layer, and multiple forward passes for stressed views, leading to higher training memory and runtime (Fig. 12 shows ~0.005s/image on A100). For edge deployment, this may be prohibitive. The paper should report FLOPs or memory usage in detail and discuss possible approximations (e.g., reducing Sinkhorn iterations, using cheaper curvature proxies) without sacrificing stability. A comparison with lightweight baselines (e.g., BN Adapt) in terms of throughput would contextualize the trade-off.

Response

We agree that runtime alone is insufficient to assess deployment cost. We therefore replaced the original accuracy–runtime comparison with a full resource profile and introduced a measured lightweight configuration, DCF-Lite.

All measurements use an NVIDIA A100, ResNet-50, and ImageNet-C under the long-horizon T-CS protocol, with batch size 64. The comparison includes BN Adapt and representative lightweight or stable baselines so that the accuracy–resource trade-off can be evaluated under a consistent setup.

Selected measurements from revised Table XI

Method	FLOPs (G)	Peak memory (MB)	Latency (ms/image)	Reported accuracy (%)
No Adapt	4.1	1,834	0.50	7.19
BN Adapt	4.1	1,944	0.50	29.83
Tent	8.2	5,685	1.40	15.79
DeYO	10.0	5,895	2.10	24.18
SAR	16.4	6,936	3.00	38.90
DCF full	24.6	14,690	5.20	43.48
DCF-Lite	9.4	5,680	3.59	42.06

Relative to BN Adapt, full DCF improves the reported accuracy by 13.65 pp while incurring additional computation. For throughput context, the reciprocal of the reported per-image latency gives approximate rates of 2,000 images/s for BN Adapt, 192 images/s for full DCF, and 279 images/s for DCF-Lite. These are latency-derived throughput estimates for the A100 benchmark in Table XI.

Measured approximations

DCF-Lite reduces cost by removing the source/candidate model copies and per-sample gradient computation, retaining gradients only through the clean adaptation branch, and using a detached Fourier probe for PSR routing and transport assignment. It further reduces the Sinkhorn iteration count to $N_{\text{sk}} = 1$. These approximations target the principal memory and computation costs of the full adaptation loop.

The Lite implementation uses four approximations to achieve these reductions: (1) replacing resident `src_model`, `hidden_model`, and `ema_model` replicas with a static source-parameter buffer θ_0 , while storing reset snapshots on the CPU; (2) computing online curvature proxies FIM_t and layer-wise gates μ_t^l from the single model’s clean-branch backward pass, then applying closed-form proximal interpolation in place; (3) detaching the Fourier probe during routing (`detach_probe=True`) to avoid auxiliary computation graphs; and (4) truncating Sinkhorn to $N_{\text{sk}} = 1$. Retention is executed in place on the single online model.

On the same A100 setup, Lite reduces FLOPs from 24.6 G to 9.4 G, peak memory from 14,690 MB to 5,680 MB, and latency from 5.20 to 3.59 ms/image, achieving 42.06% accuracy—1.42 pp below full DCF and 3.16 pp above SAR. Figure 12(d) supports low Sinkhorn iteration counts across the tested regularization range. Table IX shows only 0.09/0.22 pp gains from Expected Fisher/Hutchinson proxies at an additional 1.9/6.8 ms/image, favoring the default proxy’s accuracy–cost trade-off.

Full DCF and DCF-Lite offer two measured choices along the accuracy–resource trade-off, with hardware and batch size specified for reproducibility. The reported latency and peak memory provide concrete reference values for device-specific deployment assessment.

Changes in the manuscript: Section IV-C, “Computational Overhead,” pp. 11–12; Table XI, p. 12; Table IX, p. 10; Fig. 12(d), p. 12. Table XI supersedes the runtime-only comparison cited as Fig. 12 in the original review.

R1.4 Layer retention patterns and alternative proxies

Comment

CLR uses curvature mismatch to gate retention, but the paper does not analyze which layers (early vs. late) are typically retained or pulled back. Figure 8 shows drift trajectories but not per-layer retention gates μ_t^l . Providing a heatmap of μ_t^l over time and across corruption types would reveal whether the method consistently preserves low-level (e.g., first conv) or high-level (classifier) layers. Additionally, the curvature proxy is estimated from pseudo-label gradients, which themselves may be noisy; evaluating CLR with Fisher information or Hessian approximations would be insightful.

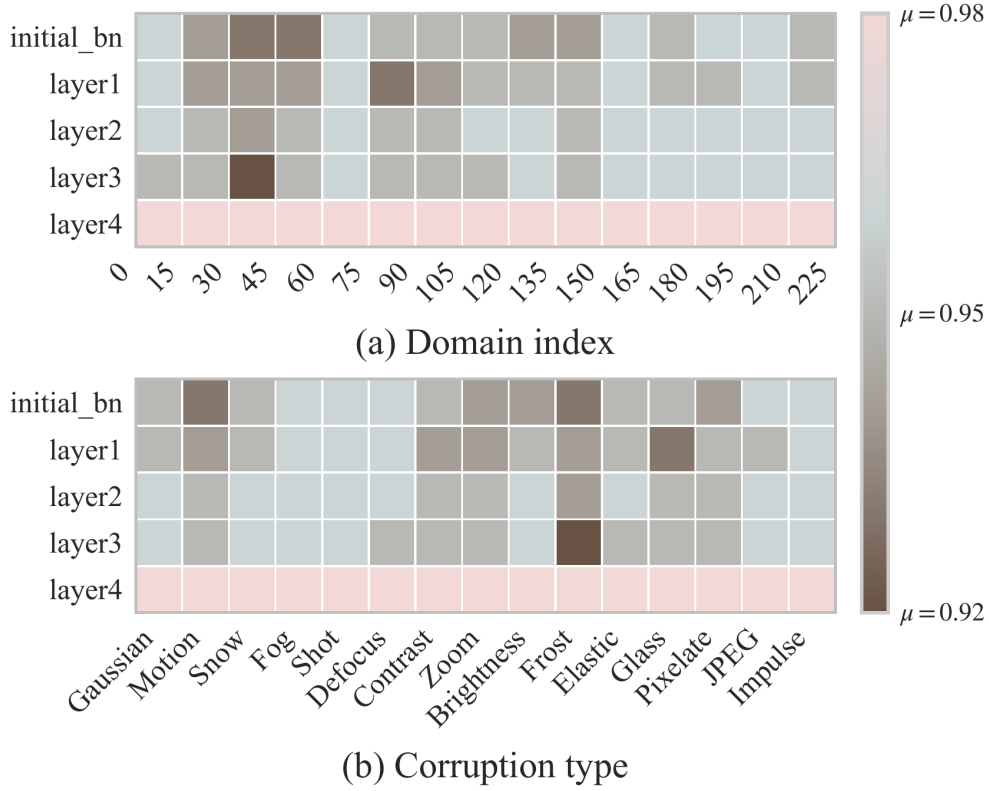
Response

We added the requested gate visualization and comparisons with alternative curvature estimators. The revised analysis separates parameter drift, shown in Fig. 8, from the retention gate itself, shown directly in the new Fig. 9.

Which groups are retained and when

CLR acts on the trainable normalization affine parameters. For ResNet, these parameters are grouped into the stem normalization group (“initial_bn”) and layer1–layer4; for ViT, grouping follows encoder blocks. These groups allow the gate patterns to be interpreted in terms of the corresponding architectural stages.

Figure 9(a) follows the gates over the long-horizon stream, and Fig. 9(b) summarizes their corruption-dependent behavior. Layer4 remains near 0.98, while the stem and layer1 generally lie around 0.93–0.96, layer2 around 0.95–0.96, and layer3 can fall to approximately 0.92 under stronger mismatch. Motion most strongly suppresses the stem gate, Glass affects layer1, and Frost produces the strongest reduction in layer3. The transient suppression around domain indices 30–45 subsequently recovers toward 0.95–0.96, showing that adaptive source anchoring preserves plasticity after transient mismatch.



Reproduced from revised Fig. 9. Retention gates over domain index (top) and corruption type (bottom).

A larger gate favors the candidate update and a smaller gate strengthens source anchoring. In the barycentric solution in Eq. (18), the gate and proxy vectors jointly determine the effective retention weights. The observed pattern therefore reflects the measured source–candidate mismatch within each layer group.

Granularity and curvature proxy comparisons

Evidence from revised Table IX

Design choice	Reported accuracy (%)	Extra latency (ms/image)
Per-normalization-layer control	42.92	0
Per-layer-block control (default)	43.48	0
Per-sample gradient-square proxy (default)	43.48	0
Expected Fisher proxy	43.57	1.9
Hutchinson-based diagonal approximation	43.70	6.8

The group comparison favors layer-block control by 0.56 pp. Expected Fisher and the Hutchinson-based approximation reach 43.57% and 43.70%, compared with 43.48% for the default gradient-square proxy, at an additional 1.9 and 6.8 ms/image. The comparison evaluates the alternative estimators under the same online pseudo-label supervision and supports the default gradient-square proxy for its accuracy–cost balance.

Section III-E further clarifies that the candidate and source proxies are computed on the same current unlabeled batch at their respective parameter states, with squared per-sample gradients averaged within each layer group. Together, this clarification and the Expected Fisher and Hutchinson comparisons show how CLR behaves with different curvature estimators.

Changes in the manuscript: Section III-E, pp. 5–6, Eqs. (17)–(20); Section IV-A, “Models and Implementation Details,” p. 6; Section IV-C, “Analysis of Curvature-aware Layer Retention,” Fig. 9 and Table IX, p. 10.

R1.5 Additional literature on uncertainty aware modeling

Comment

The literature review part can be improved by including more recent publications on machine learning methods, which can refer to Uncertainty-aware, high-precision multi-step prediction of structural health monitoring sensor streams under extreme typhoon events: an enhanced Bayesian dynamic linear model leveraging the kernel regression basis function for severe environmental adaptation, Bayesian Network in Structural Health Monitoring: Theoretical Background and Applications Review, Towards high-accuracy data modelling, uncertainty quantification and correlation analysis for SHM measurements during typhoon events using an improved most likely heteroscedastic Gaussian process.

Response

Thank you for suggesting these studies. We have incorporated all three references into Section II as cross-disciplinary work on uncertainty-aware modeling under evolving environmental conditions. The added references are:

[19] Q.-A. Wang et al., “Uncertainty-Aware, High-Precision Multi-Step Prediction of Structural Health Monitoring Sensor Streams under Extreme Typhoon Events: An Enhanced Bayesian Dynamic Linear Model Leveraging the Kernel Regression Basis Function for Severe Environmental Adaptation,” *Measurement*, vol. 262, article 120050, 2026.

[20] Q.-A. Wang, A.-W. Lu, Y.-Q. Ni, J.-F. Wang, and Z.-G. Ma, “Bayesian Network in Structural Health Monitoring: Theoretical Background and Applications Review,” *Sensors*, vol. 25, no. 12, article 3577, 2025.

[21] Q.-A. Wang et al., “Towards High-Accuracy Data Modelling, Uncertainty Quantification and Correlation Analysis for SHM Measurements during Typhoon Events Using an Improved Most Likely Heteroscedastic Gaussian Process,” *Smart Structures and Systems*, vol. 32, no. 4, pp. 267–279, 2023.

These studies contribute Bayesian dynamic modeling, Bayesian-network reasoning, and heteroscedastic uncertainty quantification for non-stationary structural-health-monitoring streams. We added them to broaden the discussion of uncertainty under evolving environments. DCF addresses the corresponding online adaptation problem in visual recognition, where the model itself is updated from unlabeled, temporally correlated target observations; its contribution is the joint control of target evidence, routed-away geometry, and layer-wise update retention.

Selected revised text

“Beyond visual tasks, uncertainty-aware modeling has also been explored to handle evolving environmental variations in non-stationary sensor streams [19]–[21].”

Changes in the manuscript: Section II, “Adaptation without Target Data,” p. 3; references [19]–[21], p. 13.

R1.6 Variability, significance, and transfer beyond ImageNet

Comment

In Figure 10, DCF shows positive gains across all source domains, but the paper does not report the variance or statistical significance. Some cells show marginal improvements ($<1\%$), which may not be robust. The authors should include standard deviations over multiple runs and perform a significance test (e.g., paired t-test) against the best baseline. Furthermore, the cross-domain evaluation adapts on one corruption and tests on the remaining 14, but the test set still shares the same ImageNet classes; evaluating on a completely different dataset (e.g., CIFAR-10-C or a medical imaging shift) would better demonstrate generalizable representation preservation. Adding such an experiment or discussing the limitation would strengthen the paper’s claims.

Response

We agree that the cross-domain results need run-to-run variability estimates and significance tests for the aggregate improvements. We have therefore added five-run variability, paired significance tests, and a second transfer benchmark on DomainNet-126.

Variability across five runs and paired comparisons

The revised Fig. 11, replacing Fig. 10 in the original submission, reports mean $\pm$ standard deviation over five matched runs. On ImageNet-C, DCF achieves a mean gain over No Adapt of 30.30 ± 0.39 pp, compared with 27.95 ± 0.05 pp for AEA, the strongest aggregate baseline in this comparison.

Aggregate results reported in revised Section IV-C

Evaluation	DCF	Best aggregate baseline	Difference	Paired test
ImageNet-C gain over No Adapt	30.30 ± 0.39 pp	AEA / 27.95 ± 0.05 pp	+2.35 pp	$t(4) = 14.30$; $p = 1.39 \times 10^{-4}$
DomainNet-126 mean accuracy	61.40%	DeYO 60.13%	+1.27 pp	$t(4) = 12.01$; $p = 2.75 \times 10^{-4}$

Both tests use five matched runs. On ImageNet-C, the paired comparison gives $t(4) = 14.30$ and $p = 1.39 \times 10^{-4}$; on DomainNet-126, DCF achieves 61.40% mean accuracy versus 60.13% for DeYO, with $t(4) = 12.01$ and $p = 2.75 \times 10^{-4}$. The ImageNet-C row reports mean gain over No Adapt, while the DomainNet row reports mean accuracy. These metrics follow the definitions used for the respective benchmarks.

These paired tests support statistically significant aggregate improvements for DCF across matched runs. The two panels summarize different evaluation units, which is important when interpreting small gains:

Fig. 11(a): Domain-aggregated transfer on ImageNet-C. Each entry reports transfer gain after adaptation to one of the 15 corruption domains, averaged across the remaining 14 unseen corruptions without further updates. DCF exceeds the domain-wise strongest baseline in 13 of 15 domains, with advantages above 1.5 pp in 12 domains and a maximum of 3.64 pp. The small-gain or lower-performing cases are Fog (+0.58 pp over AEA), Frost (-0.24 pp versus SPA), and Impulse (-2.03 pp versus SPA). These domain-wise results describe the distribution of gains; statistical significance is evaluated by the five matched run-level aggregate comparisons reported above.

Fig. 11(b): Pairwise transfer on DomainNet-126. Each entry reports mean $\pm$ standard deviation over five runs for a specific Adaptation $\rightarrow$ Evaluation pair. DCF exceeds DeYO in all 12 pairs and yields positive gains over No Adapt throughout this matrix (0.57–12.38 pp). It achieves the highest mean gain in 10 of 12 pairs; the exceptions are $P \rightarrow R$ (0.57 versus RoTTA’s 0.67 pp) and $S \rightarrow R$ (9.21 versus RoTTA’s 9.37 pp). This consistent positive transfer, together with the significant aggregate advantage over DeYO, supports representation preservation across the evaluated artistic domains.

Transfer on a distinct dataset

To extend the evaluation beyond ImageNet corruptions, we added DomainNet-126, a separate 126-class benchmark spanning Clipart, Painting, Real, and Sketch. Models adapt on one domain and are then evaluated on the remaining unseen domains without further updates. DCF attains 61.40% mean accuracy versus 60.13% for DeYO and improves over DeYO across all evaluated adaptation-to-evaluation pairs.

Together, the experiments evaluate cross-corruption transfer within ImageNet-C and cross-domain transfer on the separate DomainNet-126 benchmark, broadening the evidence for representation preservation across synthetic and artistic shifts. Each benchmark uses its own fixed label space and source classifier; the evaluation measures post-adaptation transfer to unseen domains within these visual-recognition tasks.

Changes in the manuscript: Section IV-A, “Benchmarks,” p. 6; Section IV-C, “Cross-Domain Transfer,” and Fig. 11(a)–(b), p. 11.

Response to Reviewer 2

Thank you for the positive assessment of the route–adapt–retain framework and the long-horizon evidence. We have addressed the reproducibility, visualization, practical-guidance, and literature suggestions as follows.

R2.1 Trainable parameters and the meaning of a layer

Comment

The implementation section states that the experiments follow the official Tent setup. For clarity and reproducibility, please explicitly specify which model parameters are optimized during TTA (e.g., all network parameters, only the affine parameters of normalization layers, or another subset). It would also be helpful to clarify how a layer is defined when applying Curvature-aware Layer Retention. For convolutional architectures, does this refer to an individual normalization layer, a residual block, or a larger network stage?

Response

We now specify which parameters are optimized and how they are grouped for CLR. Following Tent, SGD updates only the affine parameters of normalization layers, with learning rate 2.5×10^{-4} and momentum 0.9; convolution kernels and classifier weights remain frozen during adaptation.

For ResNet, CLR groups these trainable affine parameters into the stem normalization group (“initial_bn”) and the layer1–layer4 architectural groups. For ViT, grouping follows encoder blocks. This definition matches the layer labels used in the new gate visualization.

We additionally compare this block-level grouping against per-normalization-layer control. Table IX reports 43.48% for layer-block control and 42.92% for per-normalization-layer control, supporting the block-level grouping used by default.

Selected revised text

“Following Tent [6], SGD optimizes only normalization affine parameters (learning rate 2.5×10^{-4} , momentum 0.9). For CLR, block groups correspond to ResNet layer blocks (stem BN (initial_bn) and layer1–layer4) or ViT encoder blocks.”

Changes in the manuscript: Section IV-A, “Models and Implementation Details,” p. 6; Section III-A, p. 3; Fig. 9 and Table IX, p. 10.

R2.2 Explicit definitions of the four routing regions

Comment

The PCS–entropy visualization in Fig. 7 is a useful addition and provides good intuition for the routing mechanism. The caption already identifies Area 1 as the low-entropy/high-PCS trusted region, but the exact conditions defining all four regions are not stated explicitly. I suggest defining Areas 1–4 and their corresponding entropy/PCS threshold conditions directly in the figure caption or in the accompanying text. This would allow readers to interpret the visualization without repeatedly referring back to the routing equations.

Response

We have explicitly defined all four regions in the sample-routing text in Section III-C, immediately after Eqs. (6)–(7). Let E denote predictive entropy, s denote PCS, and v_{Ent} and v_{PCS} denote the two thresholds. The four regions are defined by the corresponding entropy/PCS inequalities shown below.

Definitions added to revised Section III-C

Region	Entropy condition	PCS condition	Role
Area 1	$E < v_{\text{Ent}}$	$s > v_{\text{PCS}}$	Trusted
Area 2	$E \geq v_{\text{Ent}}$	$s > v_{\text{PCS}}$	Routed away
Area 3	$E < v_{\text{Ent}}$	$s \leq v_{\text{PCS}}$	Routed away
Area 4	$E \geq v_{\text{Ent}}$	$s \leq v_{\text{PCS}}$	Routed away

Only Area 1 supplies trusted consistency supervision, while Areas 2–4 form the routed-away set used for geometry repair. The explicit inequalities also highlight the role of Area 3: these samples appear confident according to entropy but exhibit insufficient stress responsiveness, precisely the regime in which entropy-only selection can retain stress-inert predictions.

Changes in the manuscript: Section III-C, p. 4, following Eqs. (6)–(7); Fig. 7, p. 9, provides the corresponding PCS–entropy diagnostic visualization.

R2.3 Practical threshold guidance and OT settings

Comment

The sensitivity analysis suggests that DCF remains effective across a reasonable range of routing hyperparameters. It would be useful to provide brief practical guidance, or a simple rule of thumb, for selecting the entropy and PCS thresholds when applying DCF to a new dataset. In addition, for reproducibility, please consider reporting the remaining implementation details of the optimal transport procedure, such as the number of Sinkhorn iterations and other relevant solver settings.

Response

We now provide a starting rule for the routing thresholds and specify the OT solver settings. For a new K -class dataset, we recommend $v_{\text{PCS}} = 0.2$ and $v_{\text{Ent}} = 0.6 \ln K$ as the default starting point, followed by only coarse adjustment if necessary. The entropy threshold scales with the maximum classification entropy, $\ln K$, while the PCS threshold controls the required probability response to the Fourier stress probe. For the default OT solver, the revised manuscript specifies Sinkhorn–Knopp with $N_{\text{sk}} = 3$ iterations and entropic regularization $\varepsilon_{\text{OT}} = 0.5$; Figure 12(d) additionally evaluates lower iteration counts and alternative regularization values.

Figure 12(b) shows a broad high-accuracy region around this default setting, supporting its use as a practical initialization across datasets. The routing-ratio analysis in Table VIII further explains the trade-off between trusted-set selectivity and the amount of reliable supervision available to the online update.

Implementation settings specified in the revised manuscript

OT or routing quantity	Value or specification
Sinkhorn–Knopp iterations	$N_{\text{sk}} = 3$
Entropic regularization	$\varepsilon_{\text{OT}} = 0.5$
Transport cost	1 - cosine similarity of normalized features and centroids
Sample marginal	Uniform over the routed-away subset
Class marginal	EMA prediction prior mixed with the uniform prior
Prior EMA and uniform mixing	$\rho = 0.01$; $\alpha_b = 0.01$; uniform initialization
Centroid momentum	$\rho_M = 0.99$, followed by normalization
Gradient treatment	Transport plans and prior predictions are detached
Routing defaults	$v_{\text{PCS}} = 0.2$; $v_{\text{Ent}} = 0.6 \ln K$
Fourier amplitude default	$\lambda = 0.2$

Section III-D further specifies the shared centroids, two-view transport plans, cross-view geometry objective, empty-subset safeguards, and negligible-class-mass handling. Figure 12(d) jointly studies Sinkhorn iterations and entropic regularization. Stable downstream accuracy across the tested configurations supports efficient finite-iteration OT for online geometry repair.

Selected revised text

“Overall, DCF requires little hyperparameter tuning for stable performance.”

Changes in the manuscript: Section IV-A, “Models and Implementation Details,” p. 6; Section III-D, pp. 4–5, Eqs. (10)–(16); “Hyperparameter Sensitivity,” p. 11, and Fig. 12(b), (d), p. 12.

R2.4 Recent work on continual adaptation

Comment

Given the rapid development of continual and long-horizon TTA, the related-work discussion could potentially be strengthened by briefly acknowledging very recent studies on long-horizon stability, where relevant. For example:
[1] The Golden Subspace: Where Efficiency Meets Generalization in Continual Test-Time Adaptation, CVPR 2026.
[2] Continual Test-Time Adaptation in Computer Vision: Methods, Benchmarks, and Future Directions, TMLR 2026.

Response

Section II now cites both suggested 2026 studies and explains how their approaches relate to DCF.

[38] G. Lai, D.-W. Zhou, Z. Li, and H.-J. Ye, “The Golden Subspace: Where Efficiency Meets Generalization in Continual Test-Time Adaptation,” CVPR, 2026.

GOLD constrains updates to a classifier-aligned low-rank subspace. DCF takes a complementary approach by controlling adaptation evidence, reusing routed-away geometry, and selectively retaining candidate updates across layer groups according to source mismatch.

[28] S. K. Maharana et al., “Continual Test-Time Adaptation in Computer Vision: Methods, Benchmarks, and Future Directions,” Transactions on Machine Learning Research, 2026.

The survey provides context for teacher models, memory, reset strategies, and long-horizon stability. DCF specifically links heterogeneous target evidence to layer-wise update retention.

Selected revised text

“Unlike these approaches, DCF treats sample-side updates strictly as candidates and selectively retains them across depth according to layer-wise source mismatch.”

Changes in the manuscript: Section II, “Test-Time adaptation” and “Reliability- and Stability-aware TTA,” p. 3; references [28] and [38], p. 13.